

PURE

DIFFERENTIATION

THE PRODUCT RULE

If $u(x)$ and $v(x)$ are functions of x , then the derivative of the product $u(x) \times v(x)$ is given by:

$$u \frac{dv}{dx} + v \frac{du}{dx} \quad \text{or} \quad vu' + uv'$$

Example 1

Given that $y = \underbrace{(2x+3)}_u \underbrace{(x^2-5)}_v$, find $\frac{dy}{dx}$ using the product rule ^{deriv of product}

Solution

$$\begin{aligned} & v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx} \\ \frac{dy}{dx} &= (x^2-5)(2) + (2x+3)(2x) \\ &= 2x^2 - 10 + 4x^2 + 6x \\ &= 6x^2 + 6x - 10 \end{aligned}$$

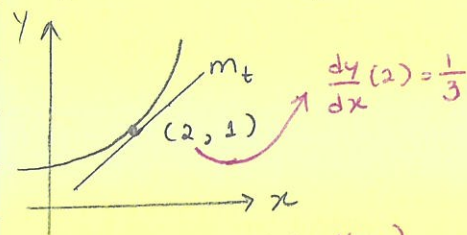
Example 2 Differentiate the following

$$y = \underbrace{20x}_u \underbrace{(x-1)^6}_v$$

Solution

$$\begin{aligned} \frac{dy}{dx} &= v \frac{du}{dx} + u \frac{dv}{dx} \\ &= (x-1)^6 \cdot 20 + 20x(6(x-1)^5) \\ &= 20(x-1)^6 + 120x(x-1)^5 \\ &= (x-1)^5 [20(x-1) + 120x] \\ &= (x-1)^5 (140x - 20) \end{aligned}$$

Equation of tangent



$$\begin{aligned} (y-y_1) &= m(x-x_1) \\ y-1 &= \frac{1}{3}(x-2) \end{aligned}$$

$$m_{\text{normal}} = -\frac{1}{m_{\text{tangent}}}$$

← common factor $(x-1)^5$

THE QUOTIENT RULE

If $u(x)$ and $v(x)$ are functions of x , then the derivative of the product $u(x) \div v(x)$ is given by:

$$\frac{u}{v} \quad \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad \text{or} \quad \frac{vu' - uv'}{v^2}$$

Example 1:

Given that $y = \frac{3x+1}{x-2}$, find $\frac{dy}{dx}$ using the quotient rule.

Solution.

$$\begin{aligned} \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} = \frac{(x-2)(3) - (3x+1)(1)}{(x-2)^2} \\ &= \frac{3x - 6 - 3x - 1}{(x-2)^2} \\ &= \frac{-7}{(x-2)^2} \end{aligned}$$

Example 2.

find the coordinates of the turning point of the curve with equation

$$y = \frac{4x+2}{\sqrt{x}}$$

$$\text{Solution. } \frac{dy}{dx} = \frac{x^{\frac{1}{2}} \times 4 - (4x+2)(\frac{1}{2}x^{-\frac{1}{2}})}{(x^{\frac{1}{2}})^2}$$

$$u = 4x+2$$

$$u' = 4$$

$$v = x^{\frac{1}{2}}$$

$$v' = \frac{1}{2}x^{-\frac{1}{2}}$$

$$= \frac{4\sqrt{x} - (2x+1)\frac{1}{\sqrt{x}}}{x}$$

$$\text{turning point } \frac{dy}{dx} = 0 \Leftrightarrow 4\sqrt{x} - (2x+1)\frac{1}{\sqrt{x}} = 0$$

$$4x - 2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

$$\begin{aligned} y &= \frac{4(\frac{1}{2}) + 2}{\sqrt{\frac{1}{2}}} \\ &= \frac{4}{\frac{1}{\sqrt{2}}} = 4\sqrt{2} \end{aligned}$$

So, the coordinates
 $(\frac{1}{2}, 4\sqrt{2})$

DIFFERENTIATING e^x and $\ln x$

At any point on the curve $y = e^x$ the gradient is equal to the y value. This means that the derivative of e^x is e^x .

we can write this as $\frac{d}{dx}(e^x) = e^x$

" $\frac{d}{dx}$ " means "the derivative of"

At any point on the curve $y = \ln x$ the gradient is equal to the reciprocal of the x value. This means that the derivative of $\ln x$ is $\frac{1}{x}$.

we can write this as

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

Chain rule:

if $y = e^{f(x)}$ then $\frac{dy}{dx} = e^{f(x)} \cdot f'(x)$

$$y = e^{(3x)} = e^{3x} \cdot (3)$$

if $y = \ln(f(x))$ then $\frac{dy}{dx} = \frac{1}{f(x)} \cdot f'(x)$

$$y = \ln(3x+2) = \frac{1}{3x+2} \cdot (3) = \frac{3}{3x+2}$$

Summary:

$$y = ae^x \Rightarrow \frac{dy}{dx} = ae^x$$

$$y = e^{ax} \Rightarrow \frac{dy}{dx} = ae^{ax}$$

$$y = e^{f(x)} \Rightarrow \frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = a \ln x \Rightarrow \frac{dy}{dx} = \frac{a}{x}$$

$$y = \ln(ax) \Rightarrow \frac{dy}{dx} = \frac{1}{x}$$

$$y = \ln(f(x)) \Rightarrow \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$(\ln x)^2 \neq \ln x^2$$

Example.

Differentiate the following

$$1. y = e^{2x}$$

$$y' = e^{2x} (2) = 2e^{2x}$$

$$y = e^{ax}$$

$$y' = ae^{ax}$$

$$4. y = 5 \ln x$$

$$y' = \frac{5}{x}$$

□ ln □

$$2. y = \frac{3}{e^{5x}}$$

$$= 3e^{-5x}$$

□ e □

$$y' = 3 \times e^{-5x} \times (-5) = -15e^{-5x}$$

$$5. y = \ln(5x)$$

$$y' = \frac{1}{5x} \times 5$$

$$= \frac{1}{x}$$

$$6. y = \ln(x^3)$$

$$y' = \frac{1}{x^3} \times 3x^2 = \frac{3}{x}$$

or

$$y = \ln(x^3) = 3 \ln(x)$$

$$y' = \frac{3}{x}$$

$$3. y = 4e^{(x^2+7)}$$

$$y' = 4 \times e^{x^2+7} \times (2x) = 8x e^{x^2+7}$$

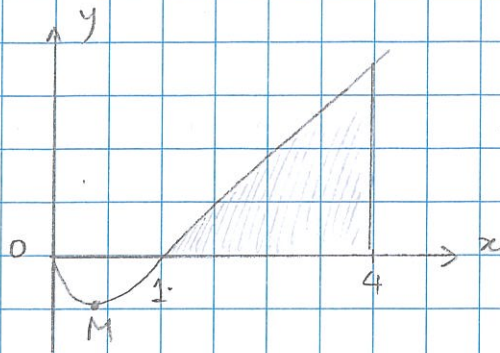
$$7. y = \ln(x^3+2)$$

$$y' = \frac{1}{x^3+2} \times 3x^2 = \frac{3x^2}{x^3+2}$$

ln [f(x)]

cannot use

log rule to simplify



The diagram shows a sketch of the curve $y = x^{\frac{1}{2}} \ln x$ and its minimum point M. The curve cuts the x-axis at the point (1, 0)

(i) Find the exact value of the x-coordinate of M [4]

$$y = x^{\frac{1}{2}} \ln(x)$$

$\begin{matrix} u & v \end{matrix}$

Product rule

$$y' = vu' + uv'$$

$$\frac{dy}{dx} = \ln(x) \times \frac{1}{2} x^{-\frac{1}{2}} + x^{\frac{1}{2}} \times \frac{1}{x}$$

M - turning point $\frac{dy}{dx} = 0$

$$\ln(x) \times \frac{1}{2\sqrt{x}} + \sqrt{x} \times \frac{1}{x} = 0$$

$\times \sqrt{x}$

$$\frac{1}{2} \ln x + x \times \frac{1}{x} = 0$$

$$\frac{1}{2} \ln x + 1 = 0$$

$$\ln x = -2$$

$$x = e^{-2} \text{ or } \frac{1}{e^2}$$

$$x = 0.1353 = 0.135$$

x-coordinate of M is 0.135

Use the quotient rule to find the x-coordinate of the stationary point of the curve.

$$y = \frac{(3-x)}{e^{2x}}$$

$$\frac{vu' - uv'}{v^2}$$

$$y' = \frac{e^{2x}(-1) - (3-x)(2)e^{2x}}{(e^{2x})^2}$$

$$y' = 0$$

$$-e^{2x} - 2e^{2x}(3-x) = 0$$

$$e^{2x}(1 + 2(3-x)) = 0$$

$$e^{2x}(7 - 2x) = 0$$

$$e^{2x} = 0 \quad \text{or} \quad 7 - 2x = 0$$

$$2x = \ln(0)$$

No solution

$$x = \frac{7}{2} = 3.5$$

$$y = \frac{(3 - \frac{7}{2})}{e^{2(\frac{7}{2})}} = \frac{-\frac{1}{2}}{e^7}$$

$$y = -\frac{1}{2e^7}$$

$\therefore$ the x-coordinate = 3.5

Differentiate the following

(review)

23.4.20

$$1. y = 3 - \frac{2}{3e^{4x}}$$

$$= 3 - \frac{2}{3}e^{-4x}$$

$$\frac{dy}{dx} = -\frac{2}{3}(-4)e^{-4x}$$

$$= \frac{8}{3}e^{-4x}$$

$$y = xe^{4x}$$

$$\frac{dy}{dx} = e^{4x} + x e^{4x}(4)$$

$$= e^{4x}(1 + 4x)$$

$$y = \frac{\ln x}{x}$$

$$\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$2. y = \ln(2 - x^3)$$

$$\frac{dy}{dx} = \frac{1}{2 - x^3} \times (-3x^2)$$

$$= -\frac{3x^2}{2 - x^3}$$

DIFFERENTIATION OF TRIG FUNCTIONS

Summary of results for sin and cos

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

remember that these results are only valid when the angle is measured in radians, so when you are using any of the derivatives of trigonometrical functions you need to work in radians.

Differentiation of $\sin(f(x))$ or $\cos(f(x))$

By the chain rule:

$$\frac{d}{dx} \sin[f(x)] = \cos[f(x)] \times f'(x)$$

$$\frac{d}{dx} \cos[f(x)] = -\sin[f(x)] \times f'(x)$$

Example:

Differentiate:

1. $\sin(kx)$ where k is a constant

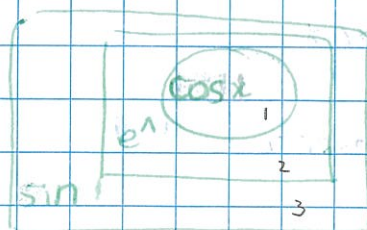
$$\begin{aligned} \frac{d}{dx}(\sin(kx)) &= [\cos(kx)] \times k \\ &= k \cos(kx) \end{aligned}$$

$$\begin{aligned} \sin(3x) &= \cos(3x) \times 3 \\ &= 3 \cos(3x) \end{aligned}$$

$$2. \cos(2x^3 - 3x)$$

$$= [-\sin(2x^3 - 3x)] \times (6x^2 - 3)$$

$$= -(6x^2 - 3)\sin(2x^3 - 3x)$$



$$3. \sin(e^{\cos x})$$

$$= [\cos(e^{\cos x})] \times [e^{\cos x}] \times [-\sin x]$$

$$= -(\sin x)(e^{\cos x})(\cos(e^{\cos x}))$$

$$y = 3 \sin x$$

(Don't use product

rule if u or v is
a constant $3\cos x$)

$$4. y = x^2 \sin x$$

Product rule

$$vu' + uv'$$

$$\frac{dy}{dx} = (\sin x)(2x) + (x^2)(\cos x)$$

$$= x [2 \sin x + x \cos x]$$

$$5. y = \tan x$$

Quotient rule

$$\frac{vu' - uv'}{v^2}$$

$$y = \frac{\sin x}{\cos x}$$

$$\frac{dy}{dx} = \frac{\cos x (\cos x) - \sin x (-\sin x)}{(\cos x)^2} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \sec^2 x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

Differentiation of sec, cosec and cot

$$f(x)$$

$$\sec x$$

$$\operatorname{cosec} x$$

$$\cot x$$

$$f'(x)$$

$$\sec x \tan x$$

$$-\operatorname{cosec} x \cot x$$

$$-\operatorname{cosec}^2 x$$

PROOF:

$$\Rightarrow y = \sec x$$

$$y = \frac{1}{\cos x}$$

$$y = (\cos x)^{-1}$$

$$\frac{dy}{dx} = (-1)(\cos x)^{-2}(-\sin x)$$

$$= \frac{\sin x}{\cos^2 x} = \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$= \tan x \sec x$$

$$\Rightarrow y = \operatorname{cosec} x$$

$$= \frac{1}{\sin x}$$

$$y = (\sin x)^{-1}$$

$$\frac{dy}{dx} = (-1)(\sin x)^{-2}(\cos x)$$

$$= -\frac{\cos x}{\sin^2 x} = -\frac{\cos x}{\sin x} \times \frac{1}{\sin x}$$

$$= -\cot x \operatorname{cosec} x$$

$$\Rightarrow y = \cot x$$

$$= \frac{\cos x}{\sin x}$$

$$\frac{dy}{dx} = \frac{(\sin x)(-\sin x) - (\cos x)(\cos x)}{(\sin x)^2}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = \frac{-1}{\sin^2 x}$$

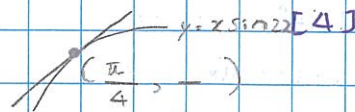
$$= -\left(\frac{1}{\sin x}\right)^2$$

$$= -\operatorname{cosec}^2 x$$

Cambridge style Questions.

EX 1.

The equation of a curve is $y = x \sin 2x$, where x is in radians.
Find the equation of the tangent to the curve at the point where $x = \frac{1}{4}\pi$



$$y = x \sin 2x$$

$$\frac{dy}{dx} = (\sin 2x)(1) + x(\cos 2x)(2)$$

$$= \sin 2x + 2x \cos 2x$$

$$\text{At } x = \frac{\pi}{4} \quad m = \sin 2\left(\frac{\pi}{4}\right) + 2\left(\frac{\pi}{4}\right)\cos 2\left(\frac{\pi}{4}\right) \quad \sin \frac{\pi}{2} = 1$$

$$= \sin\left(\frac{\pi}{2}\right) + \frac{\pi}{2} \cos\left(\frac{\pi}{2}\right) \quad \cos \frac{\pi}{2} = 0$$

$$= 1 + 0$$

$$m = 1$$

$$y = x \sin 2x$$

$$= \left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{4} = 1\left(x - \frac{\pi}{4}\right)$$

$$y = x$$

$$\frac{d}{dx} 5 \sin^2 x = \frac{d}{dx} 5(\sin x)^2$$

$$= 10 \sin x \cos x$$

$$\frac{d}{dx} 5(\sin(e^x))^2$$

$$= 10(\sin(e^x))(\cos(e^x))e^x$$

$$= 10e^x(\sin(e^x))(\cos(e^x))$$

$$\frac{d}{dx} \sin x^2 = \cos x^2 (2x)$$

EX 2. A curve has equation $y = e^{-3x} \tan x$. Find the x -coordinates of the stationary points on the curve in the interval $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$. Give your answers correct to 3 dp.

$$\text{Diff.} = 0$$

Solve for x

$$y = e^{-3x} \tan x$$

$$\frac{dy}{dx} = (\tan x)(e^{-3x} \cdot -3) + (e^{-3x})(\sec^2 x)$$

$$= e^{-3x}(-3 \tan x + \sec^2 x) = 0$$

$$e^{-3x} = 0$$

no solution

$$\text{or } -3 \tan x + \sec^2 x = 0$$

$$-3 \tan x + (\tan^2 x + 1) = 0$$

$$\tan^2 x - 3 \tan x + 1 = 0$$

$$\tan x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(1)}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

$$\tan x = \frac{3 + \sqrt{5}}{2} = 2.6180 \quad \tan x = \frac{3 - \sqrt{5}}{2} = 0.3819$$



$$x = \tan^{-1}\left(\frac{3 + \sqrt{5}}{2}\right) \\ = 1.206$$

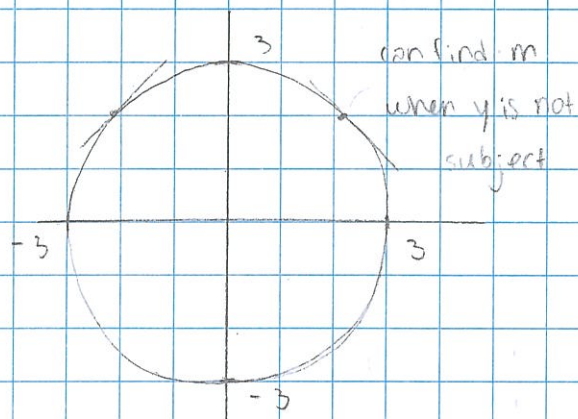
$$x = \tan^{-1}\left(\frac{3 - \sqrt{5}}{2}\right) \\ = 0.365$$

DIFFERENTIATING FUNCTIONS DEFINED IMPLICITLY

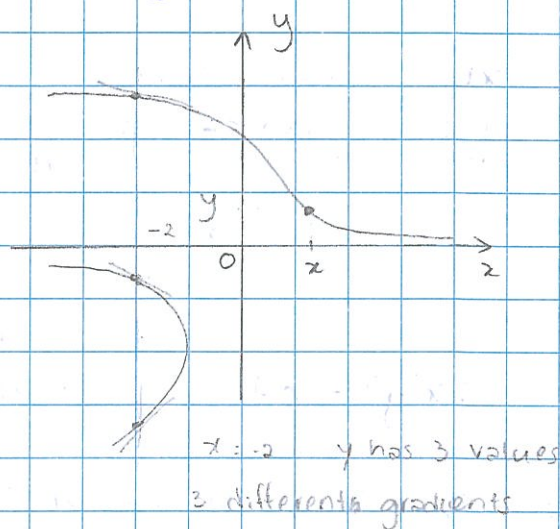
When a function is specified by an equation connecting x and y that does not have y as the subject, it is called an implicit function.

Example:

$$x^2 + y^2 = 9$$



$$y^3 + xy = 2$$



To differentiate these types of expressions, we use 'implicit differentiation' implicit = "implied"

E.g. Differentiate the following terms with respect to x .

1(a) y^2

(b) $\frac{d}{dx} (3y^4)$

$$\boxed{\frac{d}{dx} (y^2) = \frac{d}{dy} (y^2) \times \frac{dy}{dx}}$$
$$= 2y \times \frac{dy}{dx}$$

$$= 12y^3 \times \frac{dy}{dx}$$

2. $\sin y$

$$\frac{d}{dx}(\sin y) = \frac{d}{dy}(\sin y) \cdot \frac{dy}{dx} \rightarrow \text{omit (don't have to write this step)}$$

$$= \cos y \cdot \frac{dy}{dx}$$

3. $\ln y$

$$\frac{d}{dx}(\ln y) = \frac{1}{y} \cdot \frac{dy}{dx}$$

4. xy Product rule uv' $x \cdot (1 \cdot \frac{dy}{dx})$

$$\frac{d}{dx}(xy) = y(1) + x \cdot \frac{dy}{dx}$$

$$= y + x \frac{dy}{dx}$$

5. Product rule

$$3x^2y^3 - e^y = \frac{d}{dz}(3x^2y^3) - \frac{d}{dx}(e^y)$$

$$= \underbrace{y^3(6x)}_{vu' + uv'} + 3x^2(3y^2)\left(\frac{dy}{dx}\right) - e^y \cdot 1 \cdot \frac{dy}{dx}$$

$$= 6xy^3 + 9x^2y^2 \frac{dy}{dx} - e^y \frac{dy}{dx}$$

$$= 6xy^3 + \frac{dy}{dx}(9x^2y^2 - e^y)$$

The equation of a curve is given by $y^3 + xy = 2$

(i) Find an expression for $\frac{dy}{dx}$ in terms of x and y



(ii) Hence find the gradient of the curve at $(1, 1)$ and the equation of the tangent to the curve at that point.

Implicit

$$(i) \quad y^3 + xy = 2 \quad \frac{d}{dx}$$

$$3y^2 \times \frac{dy}{dx} + y(1) + x \frac{dy}{dx} = 0$$

Implicit diffⁿ

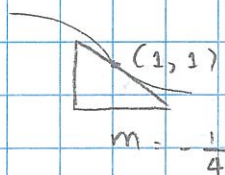
make $\frac{dy}{dx}$ the subject

$$3y^2 \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (3y^2 + x) + y = 0$$

$$\frac{dy}{dx} (3y^2 + x) = -y$$

$$\frac{dy}{dx} = \frac{-y}{3y^2 + x}$$



(ii) gradient at $(1, 1)$

$$\frac{dy}{dx} = \frac{-1}{3(1)^2 + 1} = -\frac{1}{4}$$

Eqⁿ of tangent

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{1}{4}(x - 1)$$

$$y = -\frac{1}{4}x + \frac{5}{4}$$

- (i) differentiate $x^3 + y^3 = 3xy$ with the respect to x .
 (ii) Hence find the coordinates of any stationary points.

(i)

$$x^3 + y^3 = 3xy$$

$u \quad v$
 $vu' + uv'$

$$3x^2 + 3y^2 \frac{dy}{dx} = 3y + 3x \frac{dy}{dx}$$

(ii)

S.p $\frac{dy}{dx} = 0$

$$\begin{aligned} 3x^2 + 3y^2(0) &= 3y + 3x(0) \\ 3x^2 + 0 &= 3y + 0 \\ 3x^2 &= 3y \\ x^2 &= y \end{aligned}$$

So, at S.P

$$y = x^2$$

and

$$x^3 + y^3 = 3xy$$

①

②

Sub ① into ②

$$x^3 + (x^2)^3 = 3x(x^2)$$

$$x^3 + x^6 = 3x^3$$

$$x^6 - 2x^3 = 0$$

$$x^3(x^3 - 2) = 0$$

$$x^3 = 0 \quad \text{or} \quad x^3 - 2 = 0$$

$$x = 0 \quad x = \sqrt[3]{2}$$

• Find y -values [$y = x^2$]

when $x = 0$

$$y = 0^2$$

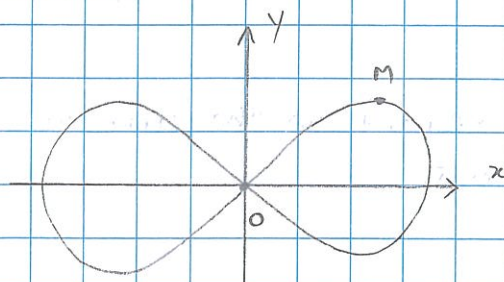
So, $(0, 0)$ is a S.P

when $x = \sqrt[3]{2}$

$$\begin{aligned} y &= (\sqrt[3]{2})^2 \\ &= \sqrt[3]{4} \end{aligned}$$

So, $(\sqrt[3]{2}, \sqrt[3]{4})$ is a S.P

Exam style question



The diagram shows the curve $(x^2+y^2)^2 = 2(x^2-y^2)$ and one of its maximum points M. Find the coordinates of M.

Find point where $\frac{dy}{dx} = 0$

$$\begin{aligned} (x^2+y^2)^2 &= 2(x^2-y^2) \\ x^4 + 2x^2y^2 + y^4 &= 2x^2 - 2y^2 \end{aligned}$$

Expand

$$4x^3 + y^2(4x) + 2x^2(2y \frac{dy}{dx}) + 4y^3 \frac{dy}{dx} = 4x - 4y \frac{dy}{dx}$$

Implicit diff.

At pt. M :

$$\begin{aligned} 4x^3 + 4xy^2 &= 4x \\ x^3 + xy^2 - x &= 0 \\ x^2 + y^2 - 1 &= 0 \end{aligned}$$

we know $x \neq 0$

At M: we know that $x^2 + y^2 = 1$ (1)
and $(x^2 + y^2)^2 = 2(x^2 - y^2)$ (2)

$y^2 = 1 - x^2$ (1)

sub (1) into (2)

$$\begin{aligned} (1)^2 &= 2(x^2 - (1 - x^2)) \\ 1 &= 2(2x^2 - 1) \\ 1 &= 4x^2 - 2 \\ x^2 &= \frac{3}{4} \end{aligned}$$

when $x^2 = \frac{3}{4}$

$$\begin{aligned} y^2 &= 1 - \frac{3}{4} = \frac{1}{4} \\ y &= \pm \sqrt{\frac{1}{4}} = \pm \frac{1}{2} \end{aligned}$$

$$x = \pm \sqrt{\frac{3}{4}} = \pm \frac{\sqrt{3}}{2}$$

need only positive

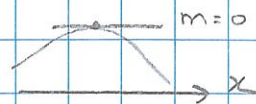
M $(\frac{\sqrt{3}}{2}, \frac{1}{2})$ or $(\frac{\sqrt{3}}{2}, -\frac{1}{2})$

(M is in the positive side)

EX 2: a curve has equation $\sin y \ln x = x - 2 \sin y$ for $-\frac{1}{2}\pi \leq y \leq \frac{1}{2}\pi$

(i) Find $\frac{dy}{dx}$ in terms of x and y .

(ii) Hence find the exact x -coordinate of the point on the curve at which the tangent is parallel to the x -axis.



$$(i) \quad \sin y \ln x = x - 2 \sin y$$

uv' u v uv'

$$(\ln x)(\cos y \frac{dy}{dx}) + \sin y (\frac{1}{x}) = 1 - 2 \cos y \frac{dy}{dx}$$

$$\ln x \cos y \frac{dy}{dx} + 2 \cos y \frac{dy}{dx} = 1 - \frac{\sin y}{x}$$

$$\frac{dy}{dx} (\ln x \cos y + 2 \cos y) = 1 - \frac{\sin y}{x}$$

$$\frac{dy}{dx} = \frac{1 - \frac{\sin y}{x}}{\ln x \cos y + 2 \cos y}$$

(ii) make $\frac{dy}{dx} = 0$ and solve

$$\frac{1 - \frac{\sin y}{x}}{\ln x \cos y + 2 \cos y} = 0$$

$$1 - \frac{\sin y}{x} = 0$$

$$\frac{\sin y}{x} = 1$$

$$x = \sin y \quad (1)$$

$$\sin y \ln x = x - 2 \sin y \quad (2)$$

$$x \ln x = x - 2x$$

$$x \ln x = -x$$

$$\ln x = -1$$

$$x = e^{-1}$$

$x = 0$

PARAMETRIC EQUATIONS

a pair of parametric equations is 2 equations where both x and y are each defined in terms of a third variable. Often t or θ are used as the third variable.

The third variable is called the parameter, and often represents the variable time.

Example:

1. $x = t^2$

$y = 2t$

2. $x = \sin \theta$

$y = \sin 2\theta$

give the shape of parabola

Graphing parametric equations:

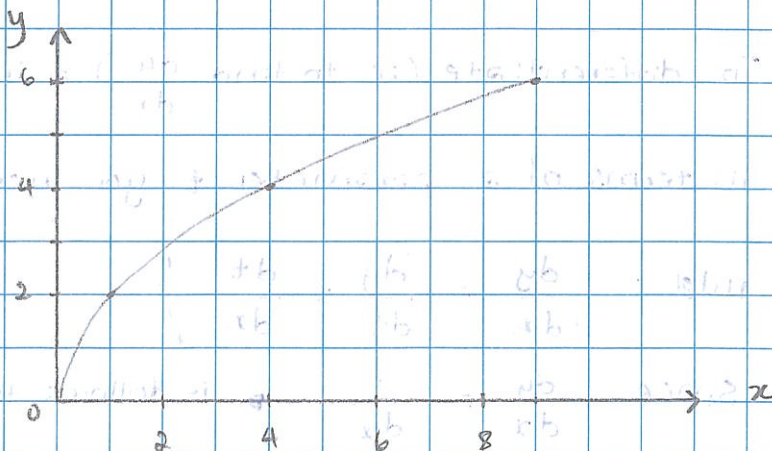
- points (x, y) can be calculated and plotted by substituting different values of t .

Example:

plot the cartesian ($x-y$) graph of the curve defined by $x = t^2$ $y = 2t$

Answer.

t	$x = t^2$	$y = 2t$
0	0	0
1	1	2
2	4	4
3	9	6



FINDING THE X-Y EQUATION BY ELIMINATING THE PARAMETER

For some pairs of parametric equations, it is possible to eliminate the parameter and obtain the cartesian equation for the curve. This is usually done by making the parameter the subject of one of the equations, and substituting this expression into the other.

Example:

Find the cartesian (x-y) equation of the curve defined by

$$x = t^2 \quad y = 2t$$

$$\downarrow$$
$$\frac{y}{2} = t$$

$$x = t^2$$

$$x = \left(\frac{y}{2}\right)^2$$

$$x = \frac{y^2}{4}$$

$$4x = y^2$$

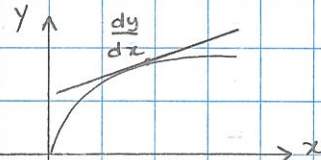


PARAMETRIC DIFFERENTIATION

To differentiate (i.e. to find $\frac{dy}{dx}$) a function which is defined

in terms of a parameter t you need to use the chain

rule: $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$



Since $\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}}$ $\rightarrow$ It follows that $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Cambridge formula sheet

If $x = f(t)$ and $y = g(t)$

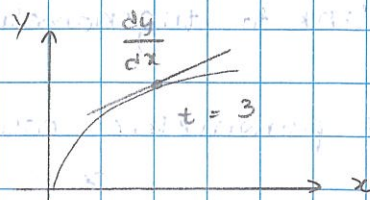
then $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$

Example:

1. Find the gradient at the point where $t = 3$ on the curve defined by $x = t^2$ $y = 2t$.

$$\frac{dx}{dt} = 2t$$

$$\frac{dy}{dt} = 2$$



$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$
$$= \frac{2}{2t} = \frac{1}{t}$$

when $t = 3$ $\frac{dy}{dx} = \frac{1}{3}$

2. Find the equation of the normal at $(-8, 4)$ to the curve which is given by the parametric equations $x = t^3$, $y = t^2$

Find the gradient of Tangent $\left(\frac{dy}{dx}\right)$

$$x = t^3$$

$$y = t^2$$



$$\frac{dx}{dt} = 3t^2$$

$$\frac{dy}{dt} = 2t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{2t}{3t^2} = \frac{2}{3t}$$

when $x = -8$

$y = 4$

t^3

t^2

$t = -2$

satisfies this condition

so, $m_N = -\frac{1}{m_T}$

$$= 3$$

Eq of normal:

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 3(x - (-8))$$

$$y - 4 = 3x + 24$$

$$y = 3x + 28$$

so, $m_T = \frac{2}{3(-2)} = -\frac{1}{3}$

Parametric equations and Parametric differentiation - exam style question

Link to trigonometry.

The parametric equations of a curve are:

$$x = t + \cos t \quad y = \ln(1 + \sin t)$$

$$\text{where } -\frac{1}{2}\pi < t < \frac{1}{2}\pi$$

(i) show that $\frac{dy}{dx} = \sec t$.

$$\frac{dx}{dt} = 1 + (-\sin t) \\ = 1 - \sin t$$

$$\frac{dy}{dt} = \frac{\cos t}{1 + \sin t}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{\cos t}{1 + \sin t}}{1 - \sin t} = \frac{\cos t}{1 - \sin^2 t} = \frac{\cos t}{1 - (1 - \cos^2 t)} \\ &= \frac{\cos t}{\cos^2 t} = \frac{1}{\cos t} = \sec t \end{aligned}$$

(ii) Hence find the x -coordinates of the points on the curve at which the gradient is equal to 3. Give your answers correct to 3 significant figures.

$$\begin{aligned} \frac{dy}{dx} &= 3 \quad \sec t = 3 \\ \frac{1}{\cos t} &= 3 \\ \cos t &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} x &= t + \cos t \\ &= 1.231 + \cos(1.231) \\ &= 1.5642 \dots \\ &= 1.56 \quad (3sf) \end{aligned}$$



$$\begin{aligned} t &= \cos^{-1}\left(\frac{1}{3}\right) \\ &= 1.231 \\ \text{or} \\ &= -1.231 \end{aligned}$$

$$\begin{aligned} \text{or} \\ x &= -1.231 + \cos(-1.231) \\ &= -0.89762 \dots \\ &= -0.898 \quad (3sf) \end{aligned}$$